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FROM CROMFORD TO CHANCERY LANE: NEW LIGHT ON THE ARKWRIGHT PATENT TRIALS

JOHN HEWISH

Court cases involving patents are useful sources for the history of technology and patent law, particularly for periods when other evidence may be scanty. Both Richard Arkwright and James Watt vigorously defended their rights in a series of trials in the latter 18th century. Central to all these cases was the legal doctrine of disclosure of the essence of an invention by means of a formal specification; this had been written into the patent grant as an absolute requirement since the first half of the 18th century.¹ Wielded by the bench, particularly as represented by Lord Mansfield, as a somewhat blunt instrument for annulling a patent, its importance was not yet fully understood by the laity of inventors. These trials show patent law developing, as well as the shortcomings of its administration, and are still cited in the courts. Many early cases were reported inadequately or not at all, but where verbatim evidence is available, as in the case of Arkwright, their value as historical sources needs no stressing.

In the 19th century the Arkwright cases were used by Guest, Ure, and others with varying emphasis, as bearing on the origins of mechanical spinning and carding.² Somewhat more recently, Wadsworth and Mann, Daniels, and others have considered Arkwright's exploitation and loss of his patents in the general context of the English textile industry.³ Indeed, nothing to do with Arkwright has been alien to us; the provocation of his attempted monopoly and the consequent organized opposition of the Manchester manufacturers are part of the image of his character and of a familiar view of early industrialization in Britain. So much is well known; yet I suggest that although much of the

MR. HEWISH is in charge of the archival collection of the Science Reference Library at the British Library.

¹A. A. Gomme, *Patents of Invention* (London, 1946), pp. 25–39.

²R. Guest, *Compendious History of the Cotton Trade . . .* (Manchester and London, 1823); A. Ure, *The Cotton Manufacture of Great Britain Investigated and Illustrated* (London, 1861), 1:266 (in part quoting Guest).

³A. P. Wadsworth and J. de L. Mann, *The Cotton Trade and Industrial Lancashire* (Manchester, 1931), pp. 491–96; G. W. Daniels, *The Early English Cotton Industry . . .* (London, 1920), chap. 4.

published evidence has been attended to, the very familiarity of these events has to some extent worked against a consideration of all the possible sources.

Here I shall not reconsider the in-court proceedings in detail or—more than briefly in passing—the wider issues but will merely add some unpublished facts concerning them from the official records, facts which I suggest modify the received view of them in minor but interesting ways.

Arkwright's defense of his patents extended over a period of some four years, from late 1781 to 1785, from his first suing for infringement to loss of his surviving unexpired patent by Crown action. It involved three separate cases: *Arkwright v. Mordaunt*, *Arkwright v. Nightingale*, and *Rex v. Arkwright*. He had taken out two patents. The first, in 1769, was for "the making of weft or yarn from cotton, flax and wool."⁴ This, the spinning patent, covered the celebrated roller-drawing principle, using pairs of rollers revolving at different speeds to replace the fingers of the hand spinner. The specification of this (in its factory setup the "water frame") was well drawn and was never directly challenged. The second patent (1775) and subject of the trials was for "certain instruments of public utility in preparing silk, cotton, flax or wool for spinning," that is, a carding machine, for producing from the fibers of cotton the light "sliver" or—eventually—the "roving" required for the spinning process.⁵ Its specification showed and described only disconnected details, some of doubtful relevance. For profitable use the inventions were interdependent: as Arkwright put it in his pamphlet plea for extension of the first patent, the carding machine was "founded in some measure on the Principles of the First Machine; these inventions, *united*, completed his great original plan."⁶

In the summer of 1781 Arkwright tried to enforce his monopoly of the carding machine by suing for infringement a number of those who, in his own words, "formed an Association of the very Persons whom he has served."⁷ Such cases between individuals were heard in the common-law courts, in this instance the Court of King's Bench, before Lord Mansfield. The defense, a formidable duo of a noted counsel, Edward Bearcroft, and Thomas (later Lord) Erskine, produced sufficient evidence of the unsatisfactory specification for the case against the first defendant, a Colonel Mordaunt, to break down. Arkwright's decision to defend his carding, rather than his still-valid spinning

⁴No. 931.

⁵No. 1,111.

⁶*The Case of Mr Richard Arkwright and Company . . .* (1782), p. 2.

⁷*Ibid.*, p. 3.

patent, is of interest; there were other means of mechanical spinning, such as the jenny, but not of carding.

This case de facto annulled his patent, and for three years he and his licensees were unprotected. In the ingeniously ghostwritten pamphlet cited above, Arkwright petitioned Parliament for recognition of the nation's debt to him by an extension of the first patent and validation of the second.⁸ This petition failed.

Late in 1784 he again commenced proceedings to reinstate the carding patent by suing a Cromford landowner, Peter Nightingale, for infringement and enticement of his workers. *Arkwright v. Nightingale* was heard in the Court of Common Pleas in February 1785, before Lord Loughborough. The defense's case rested on the specification, and Arkwright's witnesses, including James Watt and Erasmus Darwin, proved more convincing than the defense. The criterion was one that was by then established by the courts: whether a workman, experienced in the art, could make the machine from the specification.⁹

The reinstated patent did not survive for long. An association of cotton manufacturers persuaded the Crown to act against it by means of a writ of scire facias. A form of summons ("let it be made known"), this was the instrument occasionally used for official annulment of a patent. Letters patent in England, being a form of grant under the royal prerogative and sealed with the Great Seal, could be formally canceled only by Crown action.

Rex v. Arkwright was heard in the Court of King's Bench (sitting in its traditional location, Westminster Hall) before Francis Buller in June 1785. After hearing many witnesses on both sides, the judge summed up against both the novelty of the carding patent and the specification, and the jury decided for the Crown. Both the Nightingale case and *Rex v. Arkwright* were published verbatim.¹⁰

Before the 19th-century reforms—some would say long after them—the English patent system was notoriously complex, and the records reflect this. Grants under the Great Seal involved the Chancery (the royal secretariat) in its administrative role.¹¹ Patent litigation, however, could involve both administrative and judicial functions of the Chancery, the common-law courts, Parliament, and the Privy Council.

⁸Ibid.

⁹For instance by Mansfield, in *Liardet v. Johnson*, concerning the patent stucco used by the Adam brothers, 1768.

¹⁰*The Trial of a Cause Instituted by Richard Pepper Arden . . .* (London, 1785); *Richard Arkwright Esquire v. Peter Nightingale* (London [?], 1785).

¹¹H. C. Maxwell-Lyte's *Historical Notes on the Great Seal* (London, 1926) is a pioneering study of the Chancery.

My involvement in an exhibition to mark the bicentenary of *Rex v. Arkwright* in 1985 (see fig. 1) provided a tempting opportunity for a search—not necessarily exhaustive—of the court and Chancery records in a well-known neighbor institution.¹² As will be seen, a somewhat clearer picture emerged of *Arkwright v. Mordaunt*. Concerning Nightingale, it has been suggested that this case was a put-up job, the litigants being in collusion to establish the patent and discourage infringers.¹³ It can, I think, be shown that this was not so. The received view of the Crown case has been that this followed from alarm at the Nightingale verdict.¹⁴ In fact it was being brought before the latter case had been heard. Last, there is some evidence of the unconventional way in which *Rex v. Arkwright* was commenced, perhaps indicative of influence at high level.



FIG. 1.—Panels from the 1985 “Rex Vs Arkwright” bicentenary exhibit at the Science Reference Library, London. (Courtesy of the Central Office of Information, London.)

¹²“Rex Vs Arkwright,” held in collaboration with the Arkwright Society at the Science Reference Library, June 1985. The Science Reference Library is the national library for the study of patents, trademarks, and designs and includes relevant holdings of the former Patent Office Library, established in 1855. The neighbor institution is the Public Record Office.

¹³R. S. Fitton and A. P. Wadsworth, *The Strutts and the Arkwrights, 1758–1830* (Manchester, 1958), p. 86; R. L. Hills, *Richard Arkwright and Cotton Spinning* (London, 1973), p. 73; Daniels (n. 3 above), p. 105.

¹⁴Daniels and most other commentators.

Arkwright v. Mordaunt, July 1781

Knowledge of this has so far been derived from brief press notices,¹⁵ from allusions by counsel and witnesses in the later trials, and from references to it in Arkwright's pamphlet already referred to. He lost through inadequacy of the specification, but it has not been clear whether this was the jury's verdict or a "nonsuit," that is, a dismissal of Arkwright's case by the judge.¹⁶

In the Court of King's Bench records, under the legal date "Monday next after three weeks of the Holy Trinity" (in 1781, July 2), there is a ruling giving the names of the ten defendants, headed by Mordaunt's.¹⁷ Here the sheriff of Middlesex (not directly—that would have been too simple—but by reference to an earlier ruling) is required to summon a special jury at the plaintiff's expense.¹⁸ Another record, a bill of complaint lodged by Arkwright in Chancery after the Nightingale case, shows that according to the plaintiff it was indeed a nonsuit, not a jury's verdict, and shows why. Arkwright is pleading for an injunction to enforce the Nightingale verdict in his favor: "And your orator, having inadvertently produced a model of the said invention in its improved state in the said Court . . . but not supposing that the said Charles Lewis Mordaunt could have set up the defence which he did and therefore not being prepared with evidence to prove that the said model did not differ from the machine described in the said drawing in any material respect, but was the same invention as that which was described in the said instrument in writing, in consequence thereof your orator was nonsuited in the said action."¹⁹ Arkwright's claim for his specification deceives us no more than it did Lord Mansfield, but there is no reason to doubt his account of the circumstances.

Arkwright v. Nightingale, February 1785

The earliest record I have found of this case is in the common pleas archives during the previous Holy Trinity term, the summer of 1784. Here Peter Nightingale "late of Leigh" is called upon to answer Richard Arkwright, who asks for damages of £40.²⁰ No details of the

¹⁵*Manchester Mercury*, July 24, 1781, refers to a jury's verdict.

¹⁶Adair, Arkwright's counsel in the Nightingale case, called it a "nonsuit" (*Richard Arkwright Esquire* . . . [n. 10 above], p. 2); Crown counsel Bearcroft, in *Rex v. Arkwright*, claimed it was a jury's verdict (*Trial of a Cause* [n. 10 above], p. 23).

¹⁷They are Mordaunt, Clegg, Barnet, Ellison "and another," Whitehead, Charlton, Brooke, Taylor "and another," Phillips, and Leemin.

¹⁸KB 125–167 (Court of King's Bench Rule Books; "Rules" here are decisions).

¹⁹C 12-136-3 (Chancery cases).

²⁰CP 40 IND 6551 (Common Pleas, Docket Books).

charges are given; for these one can refer again to the bill of complaint lodged later in April, when Arkwright was trying to enforce the favorable verdict. In this vast document, much taken up with the text of the grant, he alleges that Nightingale had built machines from his specification with the help of workers enticed from his factory and requests Nightingale's books for assessment of damages. The latter did not hurry with his long answer, lodged in June. He admits using the machines but withholds his books, claiming that the patent is invalid and subject to action for annulment.²¹ The exchange illustrates the perennial problem of patentees—that the state will not defend their patent for them.

This attempt by Arkwright to enforce his rights seems to dispose of the possibility that there was any collusion behind the Nightingale verdict. In this, incidentally, he was awarded 1 shilling in damages. We can now see why, and how, in this token award, the collusion theory may have arisen.

Rex v. Arkwright, June 1785

Here the records show fairly conclusively that the Crown case was under way before the verdict of *Arkwright v. Nightingale* was known—and that hence the latter was nugatory, as the same issue was to be tried again. They also hint, at least, at the unorthodox way in which the Crown case was started.

By the early 19th century the writ scire facias required a definite procedure: the objector to a patent had to lodge a statement at the Petty Bag Office, part of the Chancery apparatus.²² Decision to proceed was taken by the Crown via the attorney general; the writ received his signature, becoming his fiat. The crucial words follow the recital of the letters patent: "And whereas we are given to understand that the said grant is prejudicial and inconvenient to our subjects in general. . . ." The formula is reticent, and one cannot but ask, Given to understand by whom? Much has been written on the Manchester opposition to Arkwright and on the associations opposed to patents.²³ The only source to name the alleged instigators of *Rex v. Arkwright* was published fifty years later, in 1836. It credits the initiative to Robert Peel, textile printer and grandfather of the 19th-century prime minister, and it lists the firms concerned and the subscription rate.²⁴ It has been suggested

²¹C 12-136-3, "*The Answer of Peter Nightingale. . .*"

²²T. C. Foster, *A Treatise on the Writ Scire Facias* . . . (London, 1851), p. 235 et seq.

²³Daniels (n. 3 above), pp. 93–112.

²⁴J. Wheeler, *Manchester, a Political, Social and Commercial History* . . . (Manchester and London, 1836), pp. 521–22.

that Charles Taylor, chemist, calico printer, and later secretary of the Society of Arts, prepared the case, but this also is somewhat apocryphal.²⁵ Arkwright's counsel significantly referred to "those behind the scenes, the movers and conductors of this business."

The Petty Bag Office papers throw a patchy light on the preparation of the Crown case. A formal declaration of the case (the opening statement to be read in court and embodying the text of the writ) was drawn up, dated February 12, 1785—an interesting date, since it is a few days before the Nightingale hearing.²⁶ On March 19 the Crown attorney "bespoke" a copy of the writ and paid 3 guineas for it. On April 11, payment was made for a copy of Arkwright's "privy seal bill or patent." Other payments for documents and attendances in court are dated after the trial. (There is even a note of payment to a draftsman for a copy of the specification drawing, showing that this was an official service).²⁷ In another series of records there is a petition, dated June 3 and from Arkwright's attorney, to plead *de novo*, since an earlier plea "might be considered insufficient." But in the records of this office, where it might be expected, no indication was found to confirm the identities of "those behind the scenes."²⁸

This series, however, includes two copies of the writ, one of them confirming the date, February 12.²⁹ They are in a bundle of similar documents: all the others are fiats, carrying the law officer's signature. This is curious and suggestive of unorthodox procedure. Irregular string pulling would be no more than characteristic of the age; it is recorded that Matthew Boulton dined with the judge before a crucial hearing concerning Watt's patents. On the other hand, the records may open only a narrow window on the case; they still are not necessarily complete. Some Petty Bag Office papers suggest a somewhat haphazard policy of retention. (There is a legal aphorism that "he who seeks equity must seek it with clean hands." He who seeks the records of equity may well emerge with sooty ones.) *Rex v. Arkwright* did not change patent law, but it certainly led to renewed interest in it and had wide repercussions among inventors, not least in concentrating minds on the need for reformed administration. The foregoing, I hope, has shown how legal and administrative records can improve knowledge of historically familiar events even after 200 years.

²⁵Fitton and Wadsworth (n. 13 above), p. 86n.

²⁶PRO C 206-72 (Proceedings on *Scire Facias*).

²⁷PRO C 220-2 (Petty Bag Office, daybooks).

²⁸PRO C 217-149 (Petty Bag Office, miscellaneous).

²⁹PRO C 217-152 (Petty Bag Office, miscellaneous).